Machine Learning, Chapters 1–5 Symbols and formulas

Page 1: the symbols. Then every formula, with its meaning beside it.

Symbols

Data and random variables (Ch. 2)

$x \in \mathcal{X} \subseteq \mathbb{R}^d$	one input: a vector of d features; x_j is feature j
$y \in \mathcal{Y}$	the output: a number (regression) or a class (classification)
X, Y	input and output as random variables
$\mathbb{P}(X, Y)$	joint distribution of the world (unknown)
$\mathbb{P}(Y \mid X{=}x)$	distribution of the output once the input is known
$\mathbb{E}[\cdot \mid X{=}x]$	average over that conditional distribution
$\mathcal{D} = \{(x^{(i)}, y^{(i)})\}_{i=1}^{N}$	training set: N i.i.d. examples; $x_j^{(i)}$ feature j of example i
$\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}}, \mathcal{D}_{\text{test}}$	fit θ / choose hyperparameters / final estimate, once
C; Δ^{C-1}	number of classes; probability vectors over C classes
$\mathbf{1}(\cdot)$	indicator: 1 if true, 0 otherwise

Decisions and losses (Ch. 2)

$a \in \mathcal{A}$	an action: a number, a class, "raise the alarm"
$\ell_{\text{dec}}(y, a)$	decision loss: cost of action a when the truth is y
ℓ_{train}	the loss optimised during training (often $\neq \ell_{\text{dec}}$)
$f^*(x)$	the Bayes rule: best decision when $\mathbb{P}$ is known
$m(x)$	the conditional mean $\mathbb{E}[Y \mid X{=}x]$
C_{FP}, C_{FN}	cost of a false positive / a false negative
k; $N_k(x)$	number of neighbours; the k nearest training points to x

Models, risk, training (Ch. 2–3)

f_θ, $\theta \in \Theta$	a model with parameters θ, learned from data
$\mathcal{H} = \{f_\theta\}$	hypothesis space: everything the model can represent
λ; $h \in \mathcal{C}$	a hyperparameter; a whole configuration from a candidate set
$R(\theta)$	population risk: expected loss over the world
$\widehat{R}(\theta)$, $\widehat{R}_{\text{train}}$	empirical risk: average loss on the data at hand
$\hat{\theta}$	the parameters ERM picks
$\hat{f}_{\mathcal{D}}$	the model a fixed procedure returns from dataset $\mathcal{D}$
$G(\hat{f}_{\mathcal{D}})$	generalization gap $= R(\hat{f}_{\mathcal{D}}) - \widehat{R}_{\mathcal{D}}(\hat{f}_{\mathcal{D}})$
$\mathbb{E}_{\mathcal{D}}[\cdot]$; $\bar{f}(x)$	average over training sets; average prediction at x
$\text{Bias}(x)$, $\text{Var}(x)$	systematic error; sample-to-sample spread, at x
ε, $\sigma^2(x)$	noise in Y and its variance (irreducible)
$\Omega(\theta)$	a complexity penalty ($\|\theta\|_2^2$ ridge, $\|\theta\|_1$ lasso)
$\hat{f}_h^{(-j)}$	model with config h fitted on all folds but j
K	number of folds
TP, FP, TN, FN	confusion-matrix counts

Information theory (Ch. 1)

$H(p)$; $H(p, q)$	entropy of p; cross-entropy of q relative to p
$\text{KL}(p\|q)$	Kullback–Leibler divergence, ≥ 0
$H(Y \mid X)$	conditional entropy: floor of the expected cross-entropy

Classification (Ch. 4)

$p(y{=}c \mid x)$	the posterior (true); p_θ, $\hat{p}$: a model's estimate
$s_{\theta,c}(x)$	score (logit) for class c
$\sigma(s)$	sigmoid: one score $\to$ probability in $(0, 1)$
$\text{softmax}(s)_c$	C scores $\to$ C probabilities summing to 1
$\pi_c = p(y{=}c)$	class prior
$p_\theta(x \mid y{=}c)$	class-conditional: which inputs class c produces
$\phi = (\pi, \theta)$	all parameters of a generative classifier
$\mathcal{N}(x; \mu, \Sigma)$	Gaussian density, mean μ, covariance Σ
$\hat{\mu}_c$, $\widehat{\Sigma}_c$, $\widehat{\Sigma}$	fitted class mean, class covariance, shared covariance
N_c	number of training examples of class c
n_{cj}; V	count of token j in class c; vocabulary size
θ_{cj}	probability of token j under class c (multinomial NB)
α	smoothing pseudo-count
π_t, π_d	class priors at training / at deployment
$r_t(x)$, $r_d(x)$	posterior of class 1 at training / deployment priors
τ_c	likelihood threshold of the reject option

Ensembles (Ch. 5)

M	number of models in the ensemble
$\hat{f}_m$, $\hat{p}_m$; $\hat{f}_{\text{ens}}$	model m's prediction / posterior; the ensemble's
e_m; σ^2, ρ	error of model m; its variance; correlation between two models' errors
$\mathcal{D}_m^*$	bootstrap sample m: N draws with replacement
$\text{count}(i)$; $\mathcal{M}_i$	times i was drawn; models whose sample omits i (out-of-bag)
$S_1, \ldots, S_J$; θ_j	leaves of a tree; constant predicted in leaf j
$I(S)$; $\hat{p}_c(S)$	impurity of node S; fraction of class c in it
N_t, N_L, N_R	sizes of a node and its two children
m_{try}	random candidate features per split (random forest)
Δ_t; MDI_j; I_j	credit of split t; impurity importance; permutation importance
F_m; T_m; η	boosted model after m rounds; tree m; learning rate
$r_m^{(i)}$	(pseudo-)residual of example i before round m
$F^{(i)}$; $\hat{p}^{(i)}$	current logit of example i; $\sigma(F^{(i)})$
$g^{(i)}, h^{(i)}$; G_q, H_q	1st/2nd derivative of the loss w.r.t. the logit; their sums over leaf q
$w_{q,m}$; $S_{q,m}$; Q_m	output, region and count of the leaves of tree m
λ, γ (XGBoost)	penalty on w^2; penalty per leaf
z_m; β	base-model predictions; stacking weights
$g_m(x)$, $a_m(x)$	mixture of experts: gate weight and gate score

Information theory (Chapter 1)

$H(p) = -\sum_c p_c \log p_c, \qquad H(p,q) = -\sum_c p_c \log q_c$ — entropy of p; cross-entropy of q relative to p

$\mathrm{KL}(p \,\|\, q) = \sum_c p_c \log \frac{p_c}{q_c} \;\geq 0, \qquad = 0 \iff q = p$ — Gibbs' inequality, proved with $\ln u \leq u - 1$

$H(p,q) = H(p) + \mathrm{KL}(p \,\|\, q)$ — cross-entropy is minimised exactly at $q = p$; the floor $H(p)$ is set by the problem, not by the model

Decision theory: the loss decides the action (Chapter 2)

$f^*(x) \in \arg\min_{a \in \mathcal{A}} \mathbb{E}\big[\ell_{\mathrm{dec}}(Y,a) \mid X = x\big]$ — the Bayes rule: at each x, the action with the smallest expected loss

$\mathbb{E}[\ell_{\mathrm{dec}}(Y, f^*(X))] \leq \mathbb{E}[\ell_{\mathrm{dec}}(Y, f(X))] \quad \text{for every } f$ — pointwise optimality implies global optimality

$\mathbb{E}[(Y-a)^2 \mid x] = \mathrm{Var}(Y \mid x) + \big(m(x) - a\big)^2$ — squared loss: the best a is the conditional mean $m(x) = \mathbb{E}[Y \mid x]$

absolute loss $\Rightarrow$ $a^* = \mathrm{median}(Y \mid x)$; $\quad$ pinball$_\tau$ $\Rightarrow$ quantile τ — asymmetric costs ask for a quantile

$f^*(x) = \arg\max_c \mathbb{P}(Y = c \mid X = x)$ — zero–one loss: the most probable class

declare 1 $\iff C_{\mathrm{FP}}\,(1 - p(x)) < C_{\mathrm{FN}}\,p(x)$ — unequal costs: compare the two expected costs

$\iff \; p(x) > \dfrac{C_{\mathrm{FP}}}{C_{\mathrm{FP}} + C_{\mathrm{FN}}}$ — the threshold; costly misses push it below $\frac{1}{2}$

Nearest neighbours: imitating the Bayes rule locally

$\hat{f}_k(x) = \dfrac{1}{k} \sum_{i \in N_k(x)} y^{(i)}$ — kNN regression: average the k nearest labels

$\hat{p}_k(Y = c \mid x) = \dfrac{1}{k} \sum_{i \in N_k(x)} \mathbf{1}\big(y^{(i)} = c\big)$ — kNN classification: vote among the k nearest

$d_p(x,z) = \Big(\sum_{j=1}^{d} |x_j - z_j|^p\Big)^{1/p}, \qquad d_w(x,z) = \sqrt{\sum_j w_j \Big(\dfrac{x_j - z_j}{s_j}\Big)^2}$ — Minkowski ($p = 2$ Euclidean); scaled, weighted distance: standardise first

$\mathbb{P}(\text{a neighbour within } 0.1) = 1 - \big(1 - 10^{-d}\big)^n$ — curse of dimensionality: $d = 10$, $n = 10^6$ gives $\approx 10^{-4}$; locality dies

Models, empirical risk, cross-entropy (Chapter 2)

$\mathcal{H} = \{f_\theta : \theta \in \Theta\}, \qquad \Delta^{C-1} = \Big\{p \in \mathbb{R}^C : \; p_c \geq 0, \; \sum_c p_c = 1\Big\}$ — the hypothesis space; a probabilistic classifier outputs a point of the simplex

$R(\theta) = \mathbb{E}_{(X,Y)\sim\mathbb{P}}\big[\ell_{\mathrm{train}}(Y, f_\theta(X))\big]$ — population risk: what we want, unknowable

$\widehat{R}_{\mathrm{train}}(\theta) = \dfrac{1}{N} \sum_{i=1}^{N} \ell_{\mathrm{train}}\big(y^{(i)}, f_\theta(x^{(i)})\big), \qquad \hat{\theta} \in \arg\min_\theta \widehat{R}_{\mathrm{train}}(\theta)$ — empirical risk, and empirical risk minimisation

$\mathbb{E}\big[\widehat{R}_{\mathrm{train}}(\hat{\theta})\big] \;\leq\; \min_\theta R(\theta) \;\leq\; \mathbb{E}\big[R(\hat{\theta})\big]$ — training risk is optimistic after fitting: the data choose, then grade, the model

$w = \dfrac{N\sum xy - \sum x \sum y}{N\sum x^2 - (\sum x)^2}, \qquad b = \dfrac{\sum y - w \sum x}{N}$ — least squares for $y \approx wx + b$: both partial derivatives of $\widehat{R}$ set to zero

$\widehat{R}_{01}(\theta) = \dfrac{1}{N} \sum_{i=1}^{N} \mathbf{1}\Big(y^{(i)} \neq \arg\max_c s_{\theta,c}(x^{(i)})\Big)$ — zero–one empirical risk: what we care about, but flat and unoptimisable

$\ell_{\mathrm{CE}}\big(y, \hat{p}_\theta(x)\big) = -\log \hat{p}_{\theta,y}(x), \qquad \widehat{R}_{\mathrm{CE}}(\theta) = -\dfrac{1}{N} \sum_{i=1}^{N} \log \hat{p}_{\theta,y^{(i)}}(x^{(i)})$ — cross-entropy loss: minus the log-probability given to the true class

$\mathbb{E}\big[-\log \hat{p}_{\theta,Y}(x) \mid x\big] = H\big(p(x)\big) + \mathrm{KL}\,\big(p(x) \,\|\, \hat{p}_\theta(x)\big) \;\geq\; H\big(p(x)\big)$ — at each x the expected CE is minimised by the true posterior: CE trains calibrated probabilities

$R_{\mathrm{CE}}(\theta) = H(Y \mid X) + \mathbb{E}_X\big[\mathrm{KL}\,\big(p(X) \,\|\, \hat{p}_\theta(X)\big)\big]$ — CE population risk: an irreducible floor plus a model-dependent divergence

$\hat{\theta}_\lambda = \arg\min_\theta \widehat{R}_{\mathrm{train}}(\theta;\lambda), \qquad \hat{\lambda} = \arg\min_\lambda \widehat{R}_{\mathrm{val}}(\hat{\theta}_\lambda;\lambda)$ — parameters on the training set, hyperparameters on the validation set, the test set once

$\widehat{R}_{\mathrm{CV}}(\lambda) = \dfrac{1}{K}\sum_{j=1}^{K} \widehat{R}_{\mathrm{val}}^{(j)}(\lambda)$	K-fold cross-validation: every example validates once; refit on everything after

Generalization, bias and variance, regularisation (Chapter 3)

$G(\hat{f}_{\mathcal{D}}) = R(\hat{f}_{\mathcal{D}}) - \widehat{R}_{\mathcal{D}}(\hat{f}_{\mathcal{D}})$	the generalization gap of one fitted model
$\mathbb{E}_{\mathcal{D}}\big[\widehat{R}_{\mathcal{D}}(\hat{f}_{\mathcal{D}})\big] \;\le\; \min_{f\in\mathcal{H}} R(f) \;\le\; \mathbb{E}_{\mathcal{D}}\big[R(\hat{f}_{\mathcal{D}})\big]$	why the gap is positive in expectation
$\mathcal{H}_1 \subset \mathcal{H}_2 \subset \cdots \;\Rightarrow\; \min_{f\in\mathcal{H}_{m+1}} \widehat{R}_{\mathrm{train}}(f) \le \min_{f\in\mathcal{H}_m} \widehat{R}_{\mathrm{train}}(f)$	nested families: training error never increases with capacity; validation error falls, then rises
$\bar{f}(x) = \mathbb{E}_{\mathcal{D}}[\hat{f}_{\mathcal{D}}(x)], \qquad \mathrm{Bias}(x) = \bar{f}(x) - f^*(x)$	average prediction across training sets; bias is its systematic error
$\mathrm{Var}(x) = \mathbb{E}_{\mathcal{D}}\big[(\hat{f}_{\mathcal{D}}(x) - \bar{f}(x))^2\big]$	variance: how much the prediction at x moves from one training set to another
$\mathbb{E}_{\mathcal{D},\,Y\mid x}\big[(Y - \hat{f}_{\mathcal{D}}(x))^2\big] = \sigma^2(x) + \mathrm{Bias}(x)^2 + \mathrm{Var}(x)$	noise + bias^2 + variance, for $Y = f^*(X) + \varepsilon$; add and subtract f^*, then $\bar{f}$
$\hat{\theta}_\lambda \in \arg\min_\theta \Big[\widehat{R}_{\mathrm{train}}(\theta) + \lambda\,\Omega(\theta)\Big] \;\equiv\; \min_\theta \widehat{R}_{\mathrm{train}}(\theta) \;\text{ s.t. }\; \Omega(\theta) \le c$	regularised ERM is fitting inside a constrained, nested family; $\lambda\uparrow$ is $c\downarrow$
$\Omega_2(\theta) = \lVert\theta\rVert_2^2 = \sum_j \theta_j^2 \;\;\text{(ridge)}, \qquad \Omega_1(\theta) = \lVert\theta\rVert_1 = \sum_j \lvert\theta_j\rvert \;\;\text{(lasso)}$	the two standard penalties
ridge, 1-D: $\hat{w}_\lambda = \dfrac{\sum_i x^{(i)} y^{(i)}}{\sum_i (x^{(i)})^2 + N\lambda}$	L_2 shrinks smoothly toward 0 and never reaches it
lasso, 1-D: $\hat{w} = \arg\min_w \tfrac{1}{2}(w-z)^2 + \lambda\lvert w\rvert = \mathrm{sign}(z)\,\max\big(\lvert z\rvert - \lambda,\, 0\big)$	L_1 soft-thresholds: coefficients within λ of zero become exactly zero
$t^* = \arg\min_t \widehat{R}_{\mathrm{val}}(\hat{\theta}_t), \qquad \hat{\theta}_{\mathrm{ES}} = \hat{\theta}_{t^*}$	early stopping: the epoch is a hyperparameter chosen on validation data
$\mathbb{E}\Big[\min_{h\in\mathcal{C}} \widehat{R}_{\mathrm{val}}(\hat{f}_h)\Big] \;\le\; \min_{h\in\mathcal{C}} R(\hat{f}_h)$	the best of many validation scores is optimistic; hence a test set used once
$\widehat{R}_{\mathrm{CV}}(h) = \dfrac{1}{K}\sum_{j=1}^{K} \widehat{R}_{\mathrm{val}}^{(j)}\big(\hat{f}_h^{(-j)}\big), \qquad \hat{h} \in \arg\min_{h\in\mathcal{C}} \widehat{R}_{\mathrm{CV}}(h)$	selection by CV; feature selection and preprocessing redone inside every fold
$\hat{\mu}_j = \dfrac{1}{N_{\mathrm{tr}}}\sum_{i\in\mathrm{tr}} x_j^{(i)}, \qquad \hat{\sigma}_j^2 = \dfrac{1}{N_{\mathrm{tr}}}\sum_{i\in\mathrm{tr}} \big(x_j^{(i)} - \hat{\mu}_j\big)^2$	standardisation statistics come from the training part only, or the estimate leaks
$\mathrm{Recall} = \dfrac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \qquad \mathrm{Precision} = \dfrac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}$	of the positives, how many were found; of the alarms, how many were true
$\mathrm{Specificity} = \dfrac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}}, \qquad \mathrm{BalancedAcc} = \dfrac{\mathrm{Recall} + \mathrm{Specificity}}{2}$	of the negatives, how many left alone; accuracy hides minority failure, balanced accuracy does not
$F_1 = \dfrac{2PR}{P+R}, \qquad \min(P,R) \le F_1 \le \dfrac{P+R}{2}$	harmonic mean: F_1 is dragged toward the worse of P and R

Discriminative classification: model the posterior directly (Chapter 4)

$p_\theta(y=1 \mid x) = \sigma\big(s_\theta(x)\big) = \dfrac{1}{1+e^{-s_\theta(x)}}$	binary: one score, squashed by the sigmoid
$\sigma(-s) = 1 - \sigma(s), \qquad \sigma(s) > \tfrac{1}{2} \iff s > 0$	symmetry; the boundary is $s = 0$ at equal costs
$\sigma(s) > t \iff s > \log\dfrac{t}{1-t}$	a cost threshold t moves the boundary to $s = \log\frac{t}{1-t}$
$p_\theta(y=c \mid x) = \mathrm{softmax}\big(s_\theta(x)\big)_c = \dfrac{e^{s_{\theta,c}(x)}}{\sum_{k=1}^{C} e^{s_{\theta,k}(x)}}$	C scores to C probabilities; shift-invariant, so only $C-1$ contrasts matter
$\log p_\theta(y=c \mid x) = s_{\theta,c}(x) - \log\sum_{k=1}^{C} e^{s_{\theta,k}(x)}$	log-softmax; with $C=2$ softmax is the sigmoid of the score difference
$\ell_{\mathrm{CE}}^{(i)}(\theta) = -\sum_{c=1}^{C} \mathbf{1}(y^{(i)}=c)\log p_\theta(y=c \mid x^{(i)}) = -\log p_\theta(y^{(i)} \mid x^{(i)})$	one-hot cross-entropy per example

$-\log L_{\text{cond}}(\theta) = -\sum_{i=1}^{N} \log p_\theta(y^{(i)} \mid x^{(i)}) = N\,\widehat{R}_{\text{CE}}(\theta)$	minimising cross-entropy is conditional maximum likelihood
$s_c(x) = w_c^\top x + b_c \;\Rightarrow\; \{x:\ s_c(x) = s_{c'}(x)\}$ is a hyperplane	affine scores give linear boundaries; costs only shift them, never bend them

Generative classification: model the data, then apply Bayes (Chapter 4)

$Y \sim \text{Categorical}(\pi), \quad X \mid (Y=c) \sim p_\theta(\cdot \mid y=c)$	the generative story: draw a class, then draw features from that class
$p_\phi(x, y=c) = \pi_c\, p_\theta(x \mid y=c)$	the joint distribution it defines, $\phi = (\pi, \theta)$
$p_\phi(y=c \mid x) = \dfrac{\pi_c\, p_\theta(x \mid y=c)}{\sum_{k=1}^{C} \pi_k\, p_\theta(x \mid y=k)}$	Bayes' rule: compare prior-weighted likelihoods
$= \text{softmax}\big(\log \pi_c + \log p_\theta(x \mid y=c)\big)_c$	the generative score is $\log \pi_c + \log p(x \mid c)$
$-\log L(\phi) = -\sum_c N_c \log \pi_c \;-\sum_c \sum_{i:\, y^{(i)}=c} \log p_\theta(x^{(i)} \mid y=c)$	the likelihood splits: a prior part, then one term per class on that class's data
$\hat\pi_c = \dfrac{N_c}{N}, \qquad \hat\mu_c = \dfrac{1}{N_c} \sum_{i:\, y^{(i)}=c} x^{(i)}$	MLE prior: the class frequency; MLE mean: the class average
$\widehat\Sigma_c = \dfrac{1}{N_c} \sum_{i:\, y^{(i)}=c} (x^{(i)} - \hat\mu_c)(x^{(i)} - \hat\mu_c)^\top$	MLE covariance (denominator N_c); it costs $d(d+1)/2$ parameters per class
$\mathcal{N}(x; \mu, \Sigma) = \dfrac{1}{(2\pi)^{d/2}\lvert\Sigma\rvert^{1/2}} \exp\Big[-\tfrac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\Big]$	the multivariate Gaussian density
QDA: $s_c(x) = -\tfrac{1}{2}(x-\hat\mu_c)^\top \widehat\Sigma_c^{-1}(x-\hat\mu_c) - \tfrac{1}{2}\log\lvert\widehat\Sigma_c\rvert + \log\hat\pi_c$	separate covariances: quadratic scores, quadratic boundaries
LDA: $s_c(x) = x^\top \widehat\Sigma^{-1}\hat\mu_c - \tfrac{1}{2}\hat\mu_c^\top \widehat\Sigma^{-1}\hat\mu_c + \log\hat\pi_c$	shared covariance: the $x^\top\widehat\Sigma^{-1}x$ term cancels, scores are affine, boundaries are hyperplanes
Naive Bayes: $p_\theta(x \mid y=c) = \prod_{j=1}^{d} p_\theta(x_j \mid y=c)$	conditional independence of the features given the class
$s_c(x) = \log\hat\pi_c + \sum_{j=1}^{d} \log p_{\hat\theta}(x_j \mid y=c)$	the score: prior evidence plus one additive term per feature
multinomial NB: $p_\theta(x \mid y=c) \propto \prod_{j=1}^{V} \theta_{cj}^{x_j}, \qquad \hat\theta_{cj} = \dfrac{n_{cj}+\alpha}{\sum_{\ell=1}^{V} n_{c\ell} + \alpha V}$	word counts; additive (Laplace) smoothing so an unseen word cannot zero out a class
$\hat p(x_j = v \mid y=c) = \dfrac{n_{cjv}+\alpha}{n_{cj}+\alpha K_j}, \qquad \hat\sigma_{cj}^2 \leftarrow \hat\sigma_{cj}^2 + \lambda s_j^2$	categorical NB smoothing (K_j values); Gaussian NB variance smoothing
prior shift: $p_d(y=c \mid x) = \dfrac{\pi_{d,c}\, p_{\hat\theta}(x \mid y=c)}{\sum_k \pi_{d,k}\, p_{\hat\theta}(x \mid y=k)}$	generative fix: same class-conditionals, swap in the deployment priors
$\dfrac{r_d(x)}{1-r_d(x)} = \dfrac{\pi_d/(1-\pi_d)}{\pi_t/(1-\pi_t)} \cdot \dfrac{r_t(x)}{1-r_t(x)}$	discriminative fix: rescale the odds by the prior-odds ratio; needs calibrated r_t
$\hat y = \begin{cases} c^* & \text{if } \log p_{\hat\theta}(x \mid y=c^*) \ge \tau_{c^*} \\ \text{reject} & \text{otherwise} \end{cases}$	reject option: softmax confidence is relative; a likelihood check detects unfamiliar inputs

Ensembles: averaging, trees, bagging, forests (Chapter 5)

$\hat p_{\text{ens}}(y=c \mid x) = \dfrac{1}{M}\sum_{m=1}^{M} \hat p_m(y=c \mid x), \qquad \hat y = \arg\max_c \hat p_{\text{ens}}(y=c \mid x)$	soft voting averages posteriors; hard voting counts predicted classes
$\text{Var}\Big(\sum_{m=1}^{M} e_m\Big) = M\sigma^2 + M(M-1)\rho\sigma^2 \;\Rightarrow\; \text{Var}(\bar e) = \sigma^2\Big(\rho + \dfrac{1-\rho}{M}\Big)$	why averaging helps: independent errors give σ^2/M; correlated ones leave a floor $\rho\sigma^2$
$T(x) = \sum_{j=1}^{J} \theta_j\, \mathbf{1}[x \in S_j], \qquad \theta_j^* = \arg\min_\theta \sum_{i:\, x^{(i)} \in S_j} \ell\big(y^{(i)}, \theta\big)$	a tree: one constant per leaf; mean (squared), median (absolute), majority (0–1)
$I(S) = 1 - \sum_{c=1}^{C} \hat p_c(S)^2$	Gini impurity: 0 for a pure node, largest when classes are equally mixed

$$(j^*, t^*) = \arg\max_{j,t} \left[I(S) - \frac{|S_L|}{|S|} I(S_L) - \frac{|S_R|}{|S|} I(S_R) \right]$$

the greedy split: the feature and threshold with the largest impurity reduction

$$\hat{f}_{\text{bag}}(x) = \frac{1}{M} \sum_{m=1}^{M} \hat{f}_m(x), \qquad \hat{f}_m \text{ fitted on the bootstrap sample } \mathcal{D}_m^*$$

bagging: same learner, M bootstrap samples, average or vote

$$\mathbb{P}\big(\text{count}(i) = 0\big) = \left(1 - \frac{1}{N}\right)^N \xrightarrow[N\to\infty]{} e^{-1} \approx 0.368$$

each bootstrap sample holds about 63.2% of the distinct observations

$$\hat{f}_{\text{OOB}}(x^{(i)}) = \frac{1}{|\mathcal{M}_i|} \sum_{m \in \mathcal{M}_i} \hat{f}_m(x^{(i)}), \qquad |\mathcal{M}_i| \approx 0.368\, M$$

out-of-bag prediction: a free validation estimate from the models that never saw i

$$\text{forest} = \text{bagging} + m_{\text{try}} \text{ random candidates per split}, \quad m_{\text{try}} \approx \sqrt{d}$$

decorrelates the trees: ρ falls faster than σ^2 rises; $m_{\text{try}} = d$ is plain bagging

$$\Delta_t = \frac{N_t}{N} \left[I(S_t) - \frac{N_L}{N_t} I(S_L) - \frac{N_R}{N_t} I(S_R) \right]$$

credit of one split: impurity removed, weighted by the node's share of the data

$$\text{MDI}_j = \frac{1}{M} \sum_{m=1}^{M} \sum_{t \in T_m:\ \text{split}(t)=j} \Delta_t$$

mean decrease in impurity: fast, training-based, biased toward features with many split points

$$I_j = \text{error after shuffling feature } j \text{ (OOB or validation)} - \text{error before}$$

permutation importance: reliance of the fitted model; correlated features mask each other

Ensembles: boosting, XGBoost, stacking, mixture of experts (Chapter 5)

$$F_m(x) = F_{m-1}(x) + \eta\, T_m(x), \qquad F_0 = \bar{y}$$

boosting: a sum of shrunken corrections, not an average

$$L = \tfrac{1}{2} \sum_{i=1}^{N} \left(r_m^{(i)} - \eta\, T_m(x^{(i)}) \right)^2, \qquad r_m^{(i)} = y^{(i)} - F_{m-1}(x^{(i)}) = -\frac{\partial \ell}{\partial F(x^{(i)})}$$

squared loss: fit the residuals; in general fit the negative gradient (pseudo-residuals)

$$\hat{p}^{(i)} = \sigma(F^{(i)}), \qquad \ell^{(i)} = -\big[y^{(i)} \ln \hat{p}^{(i)} + (1 - y^{(i)}) \ln(1 - \hat{p}^{(i)}) \big]$$

binary classification: boost the logit F, measure with cross-entropy

$$g = \frac{\partial \ell}{\partial F} = \hat{p} - y, \qquad h = \frac{\partial^2 \ell}{\partial F^2} = \hat{p}\,(1 - \hat{p})$$

gradient and Hessian of the log-loss with respect to the logit

$$\ell(y, F + w) \approx \ell(y, F) + g\, w + \tfrac{1}{2}\, h\, w^2 \quad \Rightarrow \quad w_{q,m} = \frac{\sum_{i \in S_q} (-g^{(i)})}{\sum_{i \in S_q} h^{(i)} + \lambda}$$

second-order Taylor step per leaf (Newton); λ regularises

$$\mathcal{L}_m = \sum_{q=1}^{Q_m} \sum_{i \in S_{q,m}} \left[g^{(i)} w_q + \tfrac{1}{2} h^{(i)} w_q^2 \right] + \gamma\, Q_m + \frac{\lambda}{2} \sum_{q=1}^{Q_m} w_q^2$$

XGBoost's one objective for a whole tree: loss change, leaf-count penalty, leaf-size penalty

$$g\, w + \tfrac{1}{2}\, h\, w^2 = \tfrac{1}{2}\, h\, (r_{\text{XGB}} - w)^2 + \text{const}, \qquad r_{\text{XGB}} = -\frac{g}{h}$$

completing the square: weighted least squares on the Newton residual, weights h

$$w_q^* = -\frac{G_q}{H_q + \lambda}, \qquad \phi_q(w_q^*) = -\frac{1}{2}\, \frac{G_q^2}{H_q + \lambda}$$

best leaf output and the loss it removes, with $G_q = \sum_{i \in S_q} g^{(i)}$, $H_q = \sum_{i \in S_q} h^{(i)}$

$$\text{gain} = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{G^2}{H + \lambda} \right] - \gamma$$

split iff gain > 0: the children must remove more loss than the parent, by more than γ

$$g_k = \hat{p}_k - y_k, \qquad \frac{\partial^2 \ell}{\partial F_k\, \partial F_r} = \begin{cases} \hat{p}_k(1 - \hat{p}_k) & k = r \\ -\hat{p}_k \hat{p}_r & k \neq r \end{cases}$$

multi-class: K logits, softmax, K trees per round; the diagonal Hessian separates the classes

$$\hat{p}_{\text{stack}} = \sigma\Big(\beta_0 + \sum_{m=1}^{M} \beta_m z_m \Big), \qquad \hat{\beta} = \arg\min_{\beta} \sum_{i=1}^{N} \ell\big(y^{(i)}, \hat{p}_{\text{stack}}^{(i)} \big)$$

stacking: a meta-model on out-of-fold base predictions; in-sample z leaks

$$p(y \mid x) = \sum_{m=1}^{M} g_m(x)\, p_m(y \mid x), \qquad g(x) = \text{softmax}\,\big(a(x)\big)$$

mixture of experts: input-dependent gate weights

$$\min \ -\sum_{i=1}^{N} \log \left[\sum_{m=1}^{M} g_m(x^{(i)})\, p_m(y^{(i)} \mid x^{(i)}) \right]$$

gate and experts trained jointly on the mixture likelihood